

Cloud-Based Data Monitoring Platform (Azure)

Project goal

The aim of the project was to design and implemented a cloud-based data monitoring and analytics platform on Microsoft Azure to collect, process and visualize system performance data from a virtualized machine hosting a Minecraft server.

Resource Group: minecraft-server

Deployed and managed cloud infrastructure using **Azure Resource Groups** In the project all the components (virtual machine, logs, data collection rules, data storage and analysis) are contained in one resource group **minecraft-server**.

Virtual Machine: minecraft-vm

On the **Azure Virtual Machine, minecraft-vm** Minecraft server was deployed with version 1.21.5 Java Edition, with the necessary software previously installed.

Log Analytics Workspace: minecraft-log-analytics

Log Analytics Workspace is a central place for collecting, storing, and analyzing log data from various sources. In this case, it's used to gather metrics and logs related to the **minecraft-vm**'s performance, such as memory and CPU usage.

Data Collection Rule: Minecraft-Performance-DCR

A data collection rule defines what data is sent to Log Analytics and how frequently. In the project, the **Minecraft-Performance-DCR** rule is responsible for passing performance metrics from the **minecraft-vm** to **minecraft-log-analytics**.

Storage Account: lakegen2minecraft

An Azure storage account (in this case, **Data Lake Gen2**) provides secure and scalable data storage.

lakegen2minecraft serves as an intermediate data store, logs from **Log Analytics** are exported to it, which are then processed in **Azure Synapse**.

Azure Synapse Workspace: synapse-minecraft

Azure Synapse, in the project serves as an environment for analyzing and visualizing data collected from a Minecraft server.

- Within **synapse-minecraft**, an **Apache Spark Pool** was created: **minecraftspark2**, which was used to process and merge data from JSON folders into Parquet format.
- For data visualization and exploration, Built-in SQL Pool was used, which allows for executing queries without the need to create a dedicated data warehouse.